HIOKI

**Instruction Manual**

# 3197Applications

3197Communicator
3197DataViewer

*Software Applications*
*For Model 3197 Power Quality Analyzer*

HIOKI E. E. CORPORATION

# Overview

## Product Overview

- “3197Applications” consists of a set of software programs dedicated for use with the HIOKI 3197 Power Quality Analyzer, namely “3197Communicator” and “3197DataViewer” software applications.
- “3197Communicator” offers remote operation of Model 3197 via a USB cable, as well as an interface to download recorded images and measurement data from Model 3197.
- “3197DataViewer” allows users to view and analyze recorded measurement data downloaded from Model 3197.

## Contents of this Manual

This manual contains the instructions on how to use the “3197Applications” set of software programs, and includes the following:

- Chapter 1: How to Use the 3197Communicator
- Chapter 2: How to Use the 3197DataViewer
- Chapter 3: Software Specifications

In order to make full use of this set of software programs, please refer to Chapter 7 of the Instruction Manual for Model 3197 Power Quality Analyzer for procedures on:

1. How to Install/Uninstall “3197Applications”
2. How to connect your computer to Model 3197 Power Quality Analyzer

## Transfer of Software, Copyright, and other Important Rights and Restrictions

**Copyright**

The title and copyright rights concerning the software's related documentation, such as the Instruction Manual and copies of the software, are the property of HIOKI E.E. CORPORATION.

**Conditions for use**

This software may not be bundled or packaged with any other software or other publications which are sold for profit without the express written permission of the copyright owner. Further, this product may not be reverse-engineered, decompiled, or disassembled without the express written permission of the copyright owner.

**Responsibility for use**

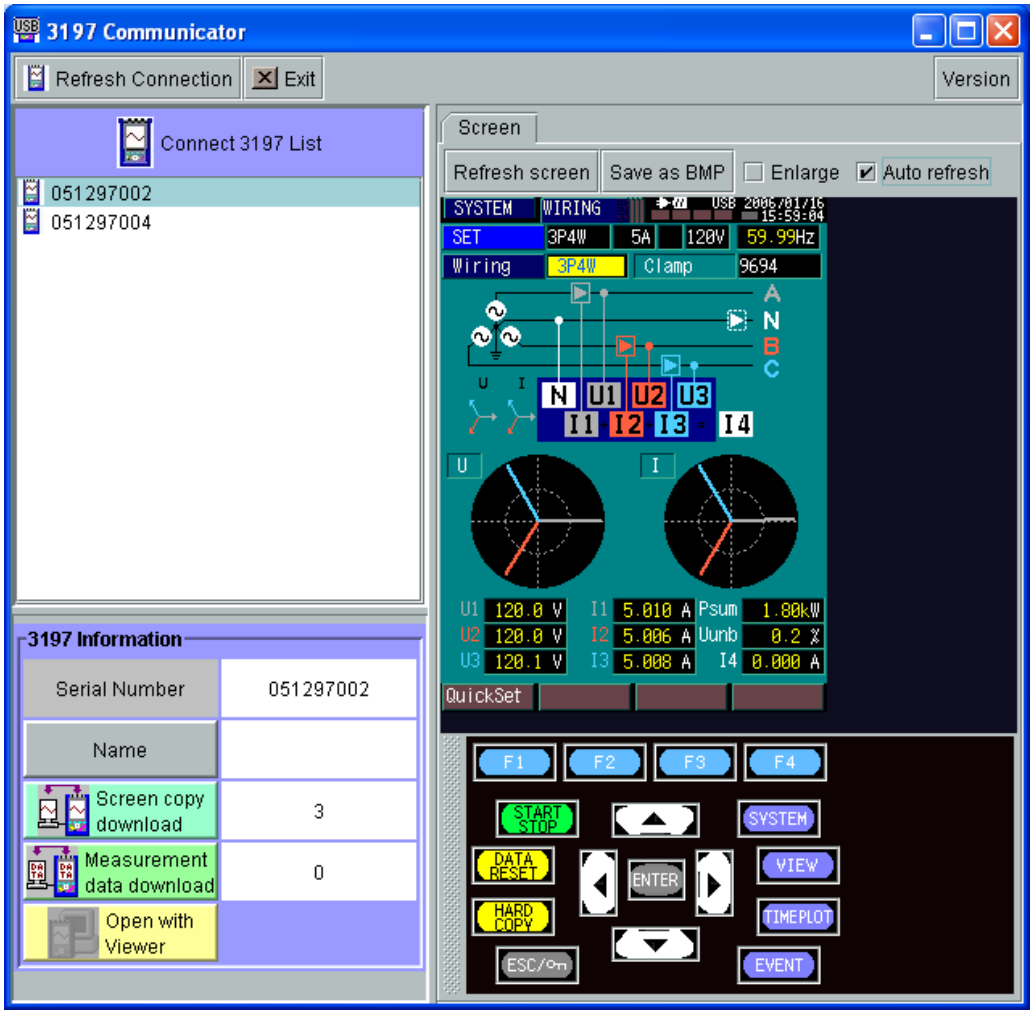
This program is provided as freeware for use at the user's risk. HIOKI E.E. Corporation disclaims any and all responsibility for any consequences arising out of use of this software, as well as for any programming errors or problems arising out of use of the software.

Windows and Microsoft Excel is a registered trademark of Microsoft Corporation

# Chapter 1: How to Use the 3197Communicator

## 1. Launching the Application

Select **Program** – **HIOKI** – **3197Applications** – **3197Communicator** from the START menu. A window similar to the one shown below will appear.




| | |
|---|---|
|   | Click “Refresh Connection” to refresh your PC’s connection with Model 3197. |
|   | Click “Exit” to quit the 3197Communicator application. |
|   | Click “Version” to check the version number of the 3197Communication application as well as that of the JAVA JRE software.   |

## 2. 3197 Connection List

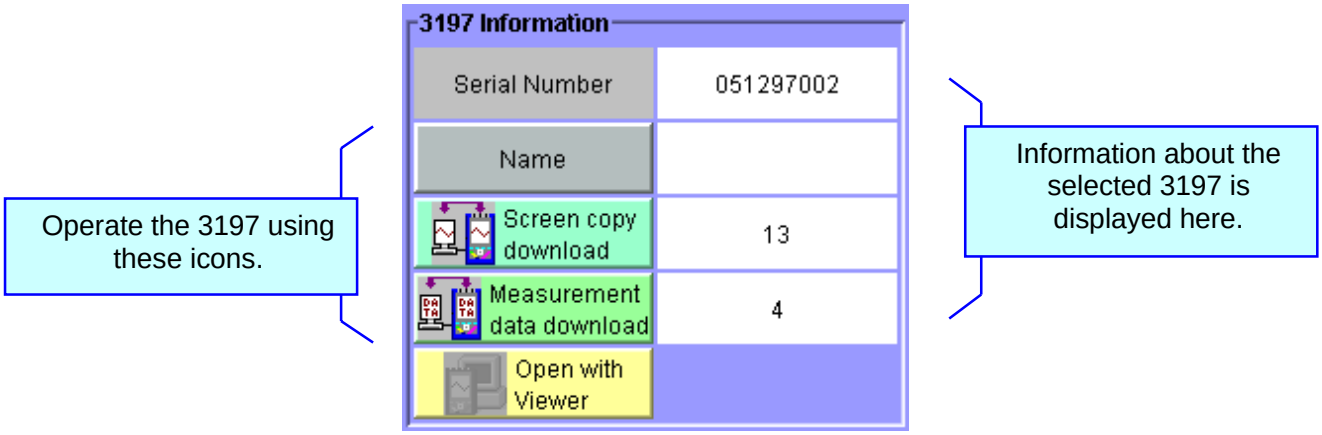
This window displays a list of Models 3197 that are connected to your PC according to serial number. In the following example, 2 units of Model 3197 are currently connected to the PC. A name for each 3197 can also be assigned by entering it in the “Name” field appearing in the “3197 Information” window. The assigned name will appear in the Connection List in place of the serial number of the 3197.




Click “Refresh Connection” to re-identify the 3197’s connected to your PC.

## 3. 3197 Information

Information for the 3197 selected in the “3197 Connection List” is displayed in this window. Operate or manage the 3197 by clicking the icons on the left column.




| | |
|---|---|
| Serial Number 051297002 | Serial number of the selected 3197. |
| Name | Assign a name to the selected 3197, which will also be loaded into the 3197’s internal memory. Up to 12 characters can be used. Even after turning off the power to the 3197, the assigned name will not be deleted. |
| Screen copy download 13 | Displays the number of saved BMP images in the selected 3197. By clicking “Screen copy download”, all the images can be downloaded to your PC. |
| Measurement data download 4 | Displays the number of measurement data saved in the selected 3197. By clicking “Measurement data download”, all saved measurement data can be downloaded to your PC. |
| Open with Viewer | Click “Open with Viewer” to open the measurement data saved in the selected 3197 the 3197DataViewer application. 3197DataViewer will launch automatically if not already open. |

Number of recorded data

## 4. Downloading Data from Model 3197

### -1. Bitmap data of screen copies

To download images of screen copies saved on the 3197, click the “Screen copy download” button. All screen shot images saved in the 3197 will be automatically saved in a new folder under the designated directory on your PC. The folder will be named “IMAGEYMMDD00” to “IMAGEYMMDD99” such that Y=last digit of the year, MM=month, and DD=day. Screen copies will be saved in BMP format, and can be viewed using your PC’s image viewer.

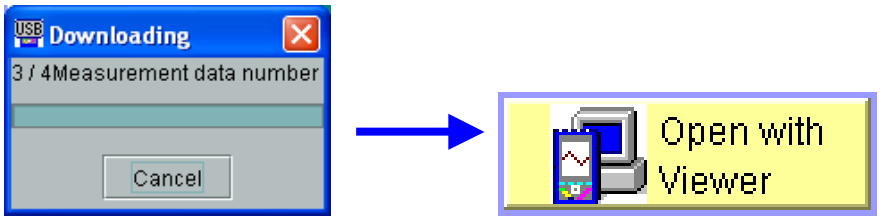
### -2. Measurement data

To download measurement data saved on the 3197, click the “Measurement data download” button. All measurement results saved in the 3197 will be saved in a new folder under the designated directory on your PC. Based on how the internal memory of the 3197 was set up to save measurement data (please see example below), new folders will automatically be created with the name “BYMMDD00” to “BYMMDD99” such that Y=last digit of the year, MM=month, and DD=day, to store the data.

***Example: Saving measurement data on February 15, 2006***

| | | |
|---|---|---|
| Ex. 1 | Internal memory not partitioned and 1 set of measurement data is saved in the 3197 | A folder named B06021500 will be created and stored with data 1/1. |
| Ex. 2 | Internal memory is partitioned into 4 segments and 2 sets of measurement data are saved in the 3197 | Two folders named B06021501 and B06021502 will be created and stored with data 1/2 and 2/2, respectively. |
| Ex. 3 | Internal memory is partitioned into 4 segments and 4 sets of measurement data are saved in the 3197 | Four folders named B06021503 to B06021506 will be created and stored with data 1/4 to 4/4, respectively. |

As soon as data has been downloaded, the “Open with Viewer” button will be active. Clicking this button will launch the 3197DataViewer application. You can also choose to open the same application using the Start menu on your PC.

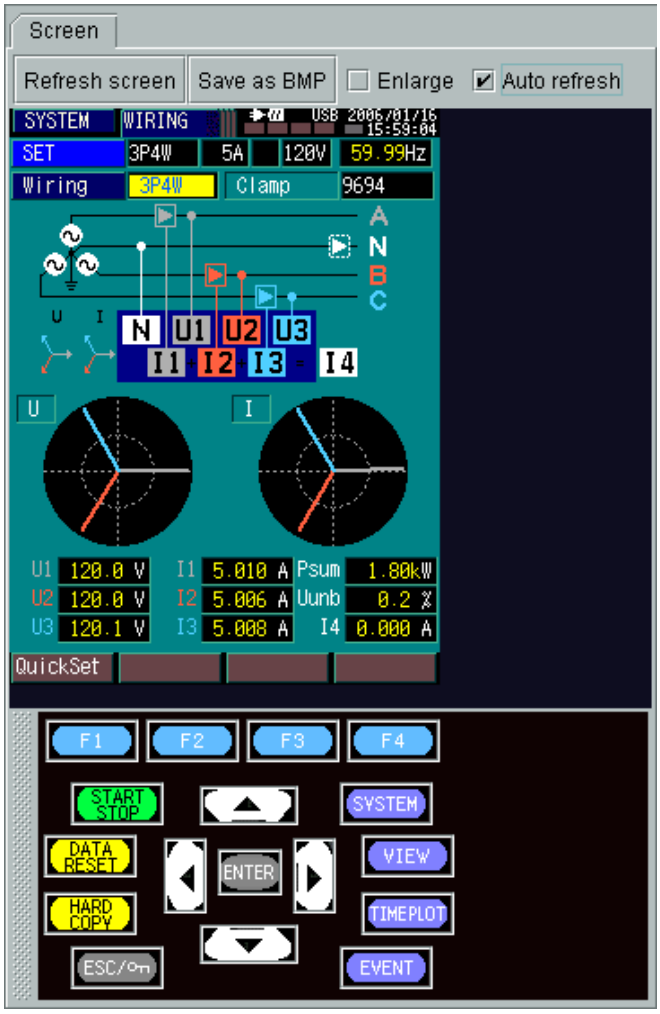



Regardless of the operating state of the 3197 PQA (whether on standby or recording) data can be downloaded at will. However, only measurement data that has been completely saved in the 3197’s internal memory can be downloaded, i.e., data currently being recorded cannot be captured.

### Definition of file names

| Measurement Data (BINARY) | Content |
|---|---|
| 3197set.set | Setting Data |
| 97intvl.itv | TIMEPLOT Data<br>(RMS, DIP/SWELL, Integrated Power) |
| 97evt000.evt to 97evt049.evt | Even Data<br>(List, waveforms) |
| 97dv000.wdu to 97dv019.wdu | Event - Voltage Fluctuation Data |
| 97inrush.inr | Event – Inrush Current Data |
| 97demand.dem | Demand Graph Data |
| 97evmoni.mon | Event Monitor Data |

## 5. Remote Control Panel, Display and Operation




**Remote Control Panel**
The screen of the selected 3197 from the 3197 Connection List will be displayed. Choose how to display the screen of the 3197 as well as perform other display functions.

**Remote Display**
The screen of the selected 3197 from the 3197 Connection List will be displayed.

**Remote Operation Keys**
Click the virtual keys with your mouse to remotely control the selected 3197 as you would on the device.

## 6. Remote Control Panel

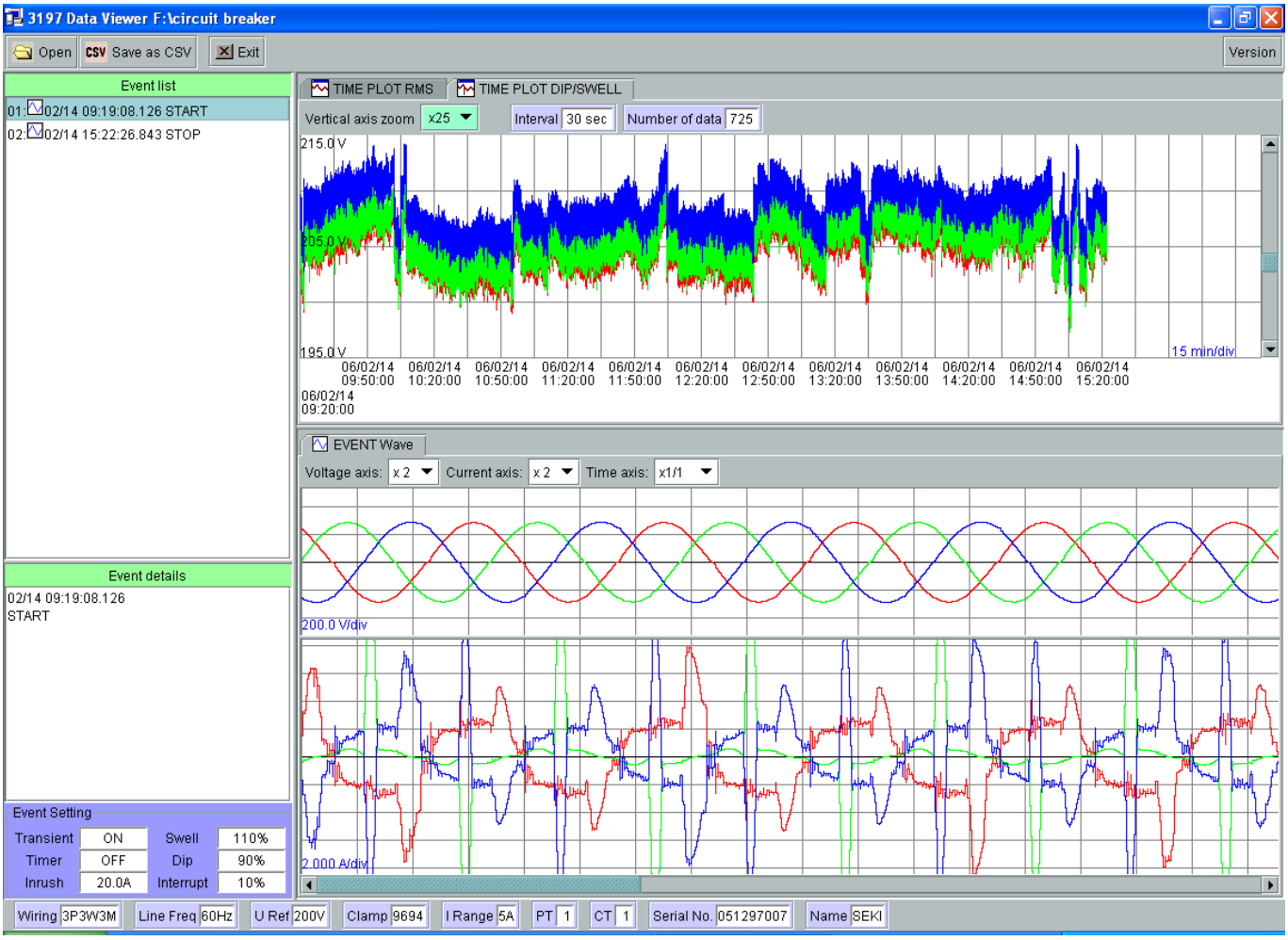


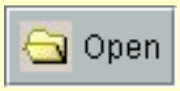
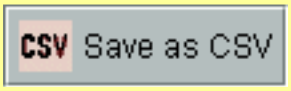

| | |
|---|---|
| Refresh screen | Click “Refresh Screen” to manually refresh the display on your PC to reflect the most current view of the selected 3197. |
| Save as BMP | Click “Save as BMP” to save a copy of the screen being displayed on the Remote Display as a BMP file on your PC. |
| ☑ Enlarge | Activate the “Enlarge” feature to see the Remote Display in 2 times its standard view. |
| ☑ Auto refresh | Activate the “Auto Refresh” feature to automatically refresh the Remote Display every second. |

# Chapter 2: How to Use the 3197DataViewer

## 1. Launching the Application

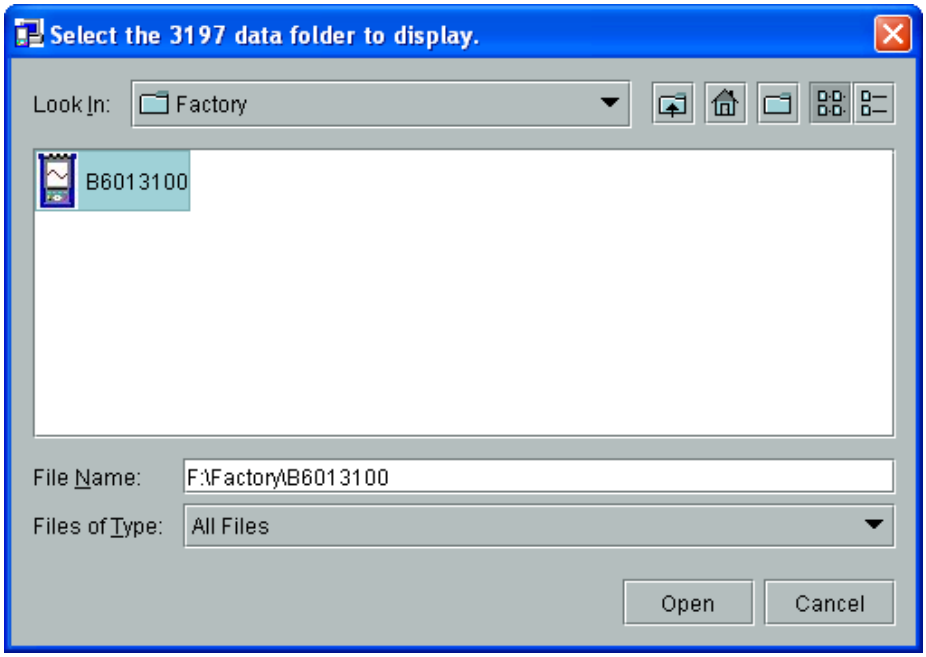
Select **Program** – **HIOKI** – **3197Applications** – **3197DataViewer** from the START menu. A window similar to the one shown below will appear.




| | |
|---|---|
|  | Click “Open” to select the folder containing the 3197 measurement data you wish to view. |
|  | Click “Save as CSV” to save the measurement data as a CSV file. |
|  | Click “Exit” to quit the 3197DataViewer application. |
|  | Click “Version” to check the version number of the 3197DataViewer application as well as that of the JAVA JRE software. |

## 2. Loading Measurement Data

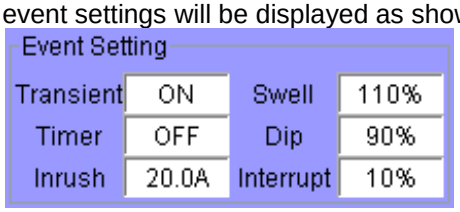
Click “Open” and a window entitled “Select the 3197 Data Folder to Display” will appear. Choose the folder containing the data you wish you view (e.g., B6020100). All available measurement data folders will appear next to an image icon of the 3197.




## 3. Settings

### -1. Event Settings Information

The event settings will be displayed as shown below.

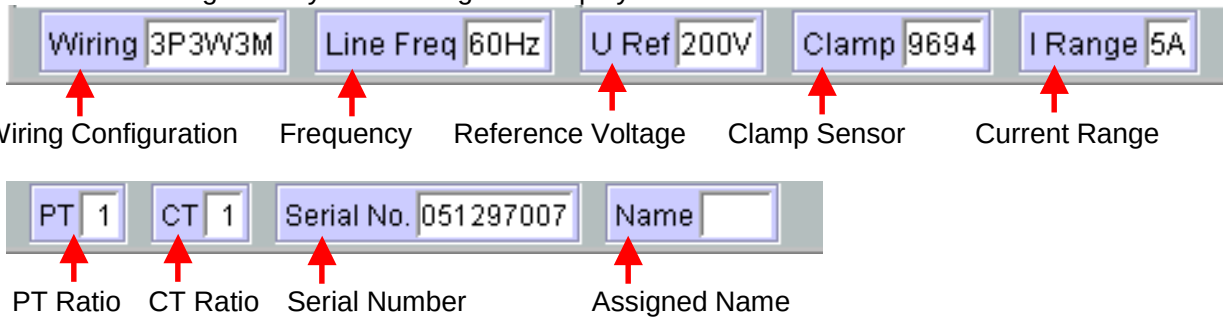



**Definitions**

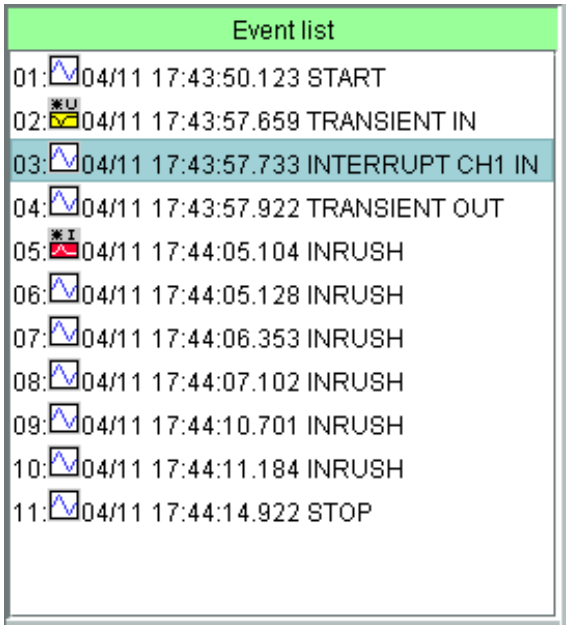
| | | | |
|---|---|---|---|
| **Transient** | Transient Overvoltage Detection ON or OFF | **Swell** | Swell Threshold |
| **Timer** | Timer ON or OFF | **Dip** | Dip Threshold |
| **Inrush** | Inrush Current Threshold | **Interrupt** | Sudden Interruption Threshold |

### -2. Measurement Settings Information

Measurement settings and system settings are displayed as shown below.



## 4. Event List




This window displays the event list.

The contents in the event list are the same as the event list on Model 3197.

| | |
|---|---|
|  | Normal event marker with a voltage/current waveform. |
|  | Event marker with event voltage fluctuation graph in addition to a voltage/current waveform. |
|  | Event marker with inrush current graph in addition to a voltage/current waveform. |

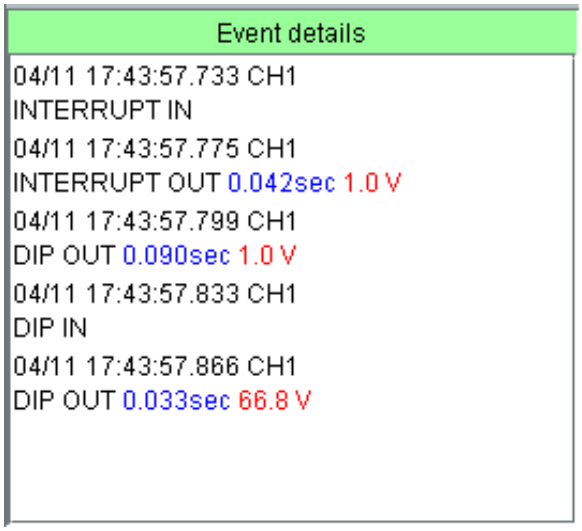
The event to view or analyze can be selected by clicking on the item on the event list. Also, up and down cursor keys can be used to select the event.

### Contents of Event List

| Item | Example | Contents | Explanation |
|---|---|---|---|
| Event number | 03 | Event number | Display from 1 to 50 |
| Date | 10/03 | Month/Year | |
| Time | 13:49:11.320 | Hour:Minute:Second | Display to milli-second units |
| Event type | START | Start | |
| | STOP | Stop | |
| | MANUAL | Manual | |
| | TIMER | Timer | |
| | SWELL | Voltage swell | |
| | DIP | Voltage dip | |
| | INTER. | Interruption | |
| | TRAN. | Transient overvoltage | |
| | INRUSH | Inrush current | |
| Channel | CH1 | Channel 1 | Displayed for voltage swell, voltage dip or interruption. |
| | CH2 | Channel 2 | |
| | CH3 | Channel 3 | |
| IN/OUT | IN | Event in | Displayed for voltage swell, voltage dip, interruption or transient overvoltage. |
| | OUT | Event out | |

When both an event voltage fluctuation graph and inrush current graph exist for a single event, the inrush current graph icon will be displayed.

## 5. Event Details




Displays the event details of the selected event in the event list.

The contents of the event details are the same as those appearing in the event details screen on Model 3197.

### Details of Event Details

| Item | Example | Contents | Explanation |
|---|---|---|---|
| Date | 03/09 | Month / day | |
| Time | 18:25:14.403 | Hour:MinuteSecond | Display to milli-second units |
| Event type | START | Start | |
| | STOP | Stop | |
| | MANUAL | Manual | |
| | TIMER | Timer | |
| | SWELL | Voltage swell | |
| | DIP | Voltage dip | |
| | INTER. | Interruption | |
| | TRAN. | Transient overvoltage | |
| | INRUSH | Inrush current | |
| Channel | CH1 | Channel 1 | Displayed for voltage swell, voltage dip and interruption. |
| | CH2 | Channel 2 | |
| | CH3 | Channel 3 | |
| IN/OUT | IN | Event in | Displayed for voltage swell, voltage dip, interruption and transient overvoltage. |
| | (omitted) | Event out | |
| Event period | 1min38.283sec | Minute:Second<br>(Displayed to milli-second units) | Displayed for voltage swell, voltage dip, interruption, and transient over voltage. |
| Voltage value | 0.3V | Maximum voltage for the voltage swell | Displayed for the event out of the voltage swell, voltage dip and interruption. |
| | | Residual voltage for the voltage dip | |
| | | Residual voltage for the interruption | |

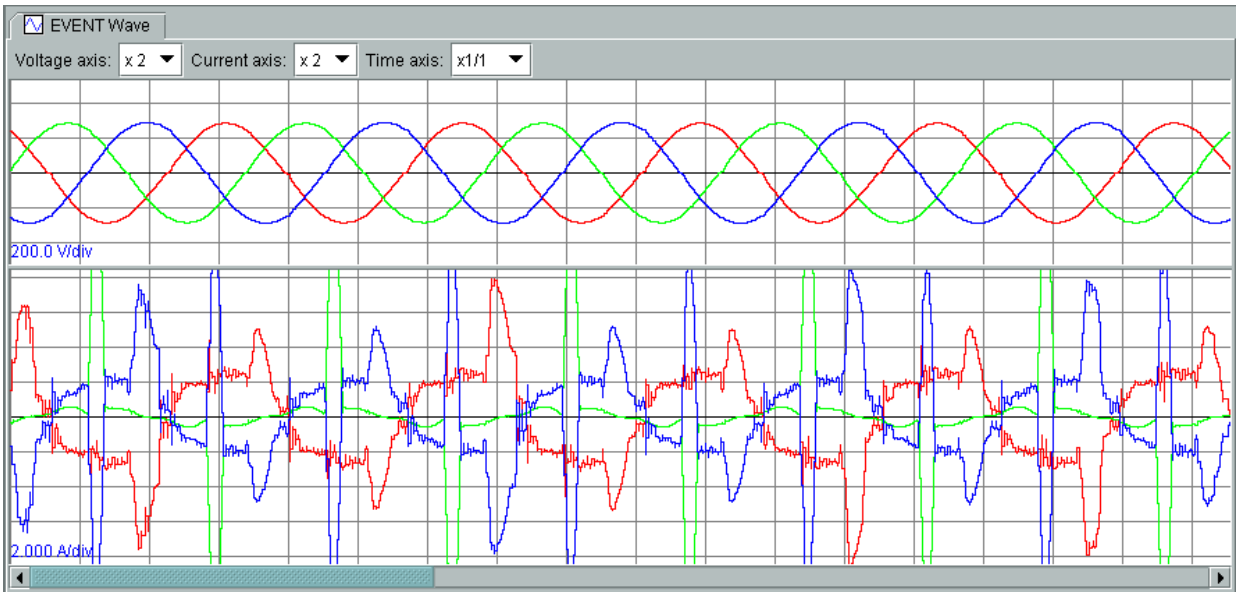
## 6. Event Contents Window

The screens of [EVENT Waveform], [EVENT Voltage Fluctuation] and [EVENT INRUSH] are switchable by clicking the respective tab.




**-1. EVENT Waveform**

In [EVENT Waveform] screen, you can analyze the waveform by changing the magnification rate of the vertical and horizontal axes. You can also scroll in the horizontal axis direction.

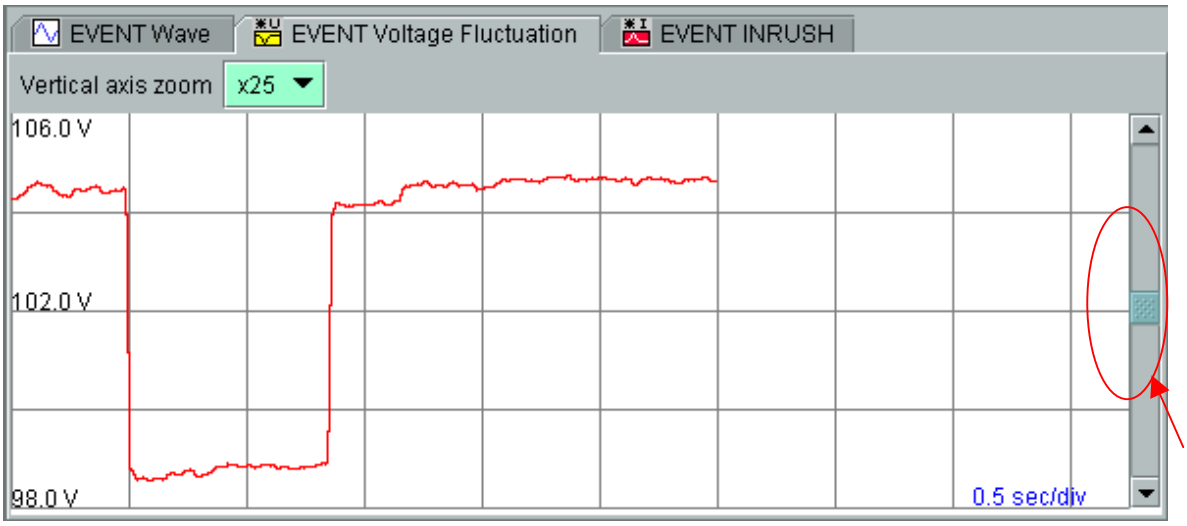



| | |
|---|---|
| ——— | Waveforms of U1 and I1 |
| ——— | Waveforms of U2 and I2 |
| ——— | Waveforms of U3 and I3 |

| | |
|---|---|
|  | Change the voltage axis magnification by selecting in x1, x2, x5, x10, x20 or x50. |
|  | Change the current axis magnification by selecting in x1, x2, x5, x10, x20 or x50. |
|  | Change the time axis magnification by selecting in x1/1. x1/2, x1/5 or x1/10. |

**-2. EVENT Voltage Fluctuation**

In [EVENT Voltage Fluctuation] screen, the horizontal axis is automatically adjusted to display the entire data period according to the window size of DataViewer. The magnification rate of the vertical axis can be changed and scrolled.

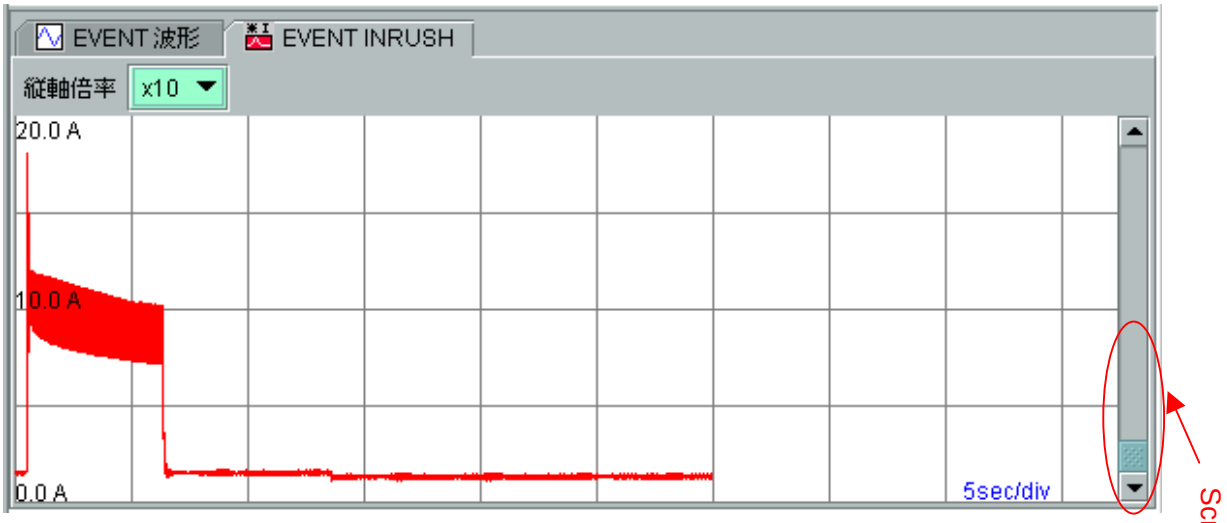



| | | | | | |
|---|---|---|---|---|---|
| ——— | Voltage RMS of U1 | ——— | Voltage RMS of U2 | ——— | Voltage RMS of U3 |

| | |
|---|---|
| Vertical axis zoom x25 | Change the vertical axis magnification rate by selecting in x1, x2, x5, x10, x25 or x50. |

**-3. EVENT INRUSH**

In the [EVENT INRUSH] screen, the horizontal axis is automatically adjusted to display the entire data period according to the window size of DataViewer. The magnification rate of the vertical axis can be changed and scrolled.




| | | | | | |
|---|---|---|---|---|---|
| ——— | Current RMS of U1 | ——— | Current RMS of U2 | ——— | Current RMS of U3 |

| | |
|---|---|
| Vertical axis zoom x25 | Change the vertical axis magnification by selecting in x1, x2, x5, x10, x25 or x50. |

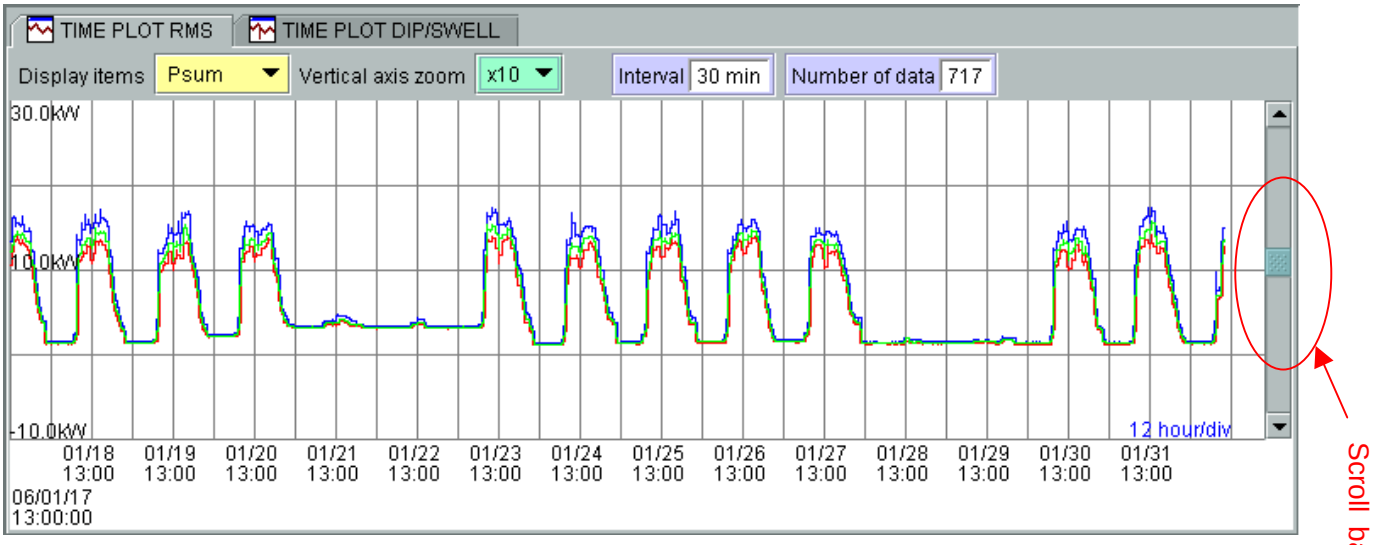
## 7. TIME PLOT Graph

The screens of the [TIME PLOT RMS] and [TIME PLOT DIP/SWELL] are switchable by clicking the tab.




### -1. TIME PLOT RMS

In [TIME PLOT RMS] screen, horizontal axis is automatically adjusted to display the entire data period according to the window size of DataViewer. The magnification rate of vertical axis can be changed and scrolled.




Maximum vale (in an interval)
Average value (in an interval)
Minimum value (in an interval)

| | |
|---|---|
|   | Select the item to display using the pull down menu. |
|   | Change the vertical axis magnification rate by selecting from x1, x2, x5, x10, x25 or x50. |
|   | Displays the recording interval setting on Model 3197. |
|   | Displays the number of data recorded by Model 3197. |

Displayable Items

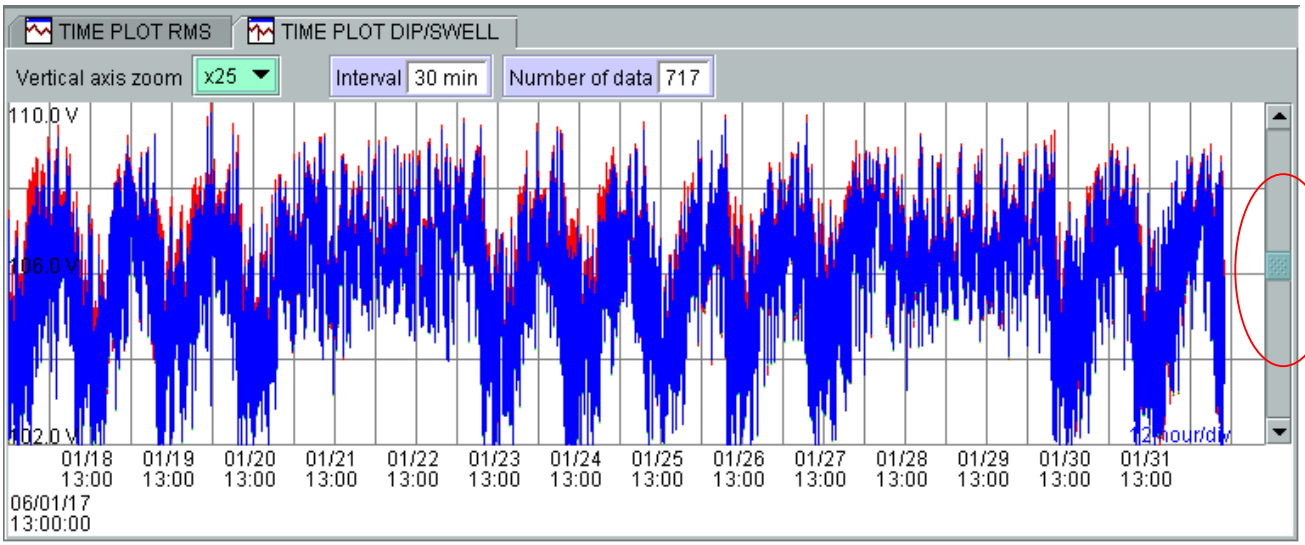
| Name | Contents |
|---|---|
| Freq | Frequency |
| U1 | CH1 Voltage |
| I1 | CH1 Current |
| U2 | CH2 Voltage |
| I2 | CH2 Current |
| U3 | CH3 Voltage |
| I3 | CH3 Current |
| I4 | CH4 Current |
| U1peak+ | Voltage Waveform Peak+ of CH1 |
| U1peak- | Voltage Waveform Peak- of CH1 |
| I1peak+ | Current Waveform Peak+ of CH1 |
| I1peak- | Current Waveform Peak- of CH1 |
| U2peak+ | Voltage Waveform Peak+ of CH2 |
| U2peak- | Voltage Waveform Peak- of CH2 |
| I2peak+ | Current Waveform Peak+ of CH2 |
| I2peak- | Current Waveform Peak- of CH1 |

| Name | Contents |
|---|---|
| U3peak+ | Voltage Waveform Peak+ of CH3 |
| U3peak- | Voltage Waveform Peak- of CH3 |
| I3peak+ | Current Waveform Peak+ of CH3 |
| I3peak- | Current Waveform Peak- of CH3 |
| Uave | Average Voltage |
| Iave | Average Current |
| Psum | Total Active Power for 3 phases |
| Qsum | Total Reactive Power for 3 phases |
| Ssum | Total Apparent Power for 3 phases |
| PFsum | Total Power Factor for 3 phases |
| DPFsum | Total Displaced Power Factor for 3 phases |
| THD1 | Total Harmonic Voltage Distortion Ratio for CH1 |
| THD2 | Total Harmonic Voltage Distortion Ratio for CH2 |
| THD3 | Total Harmonic Voltage Distortion Ratio for CH3 |
| Uunb | Voltage Unbalance Ratio |

*Some items may not be available depending on the actual wiring condition.*

**-2. TIME PLOT DIP/SWELL**

In [TIME PLOT DIP/SWELL] screen, the horizontal axis is automatically adjusted to display the entire data period according to the window size of data viewer. The magnification rate of vertical axis can be changed and scrolled.



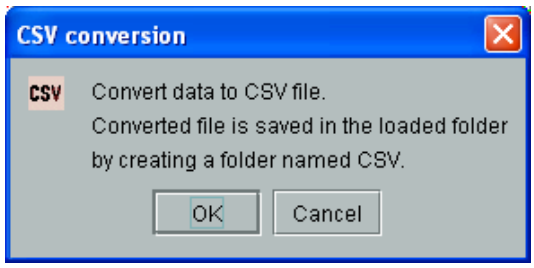

| | |
|---|---|
| ——— | Voltage RMS of U1 |
| ——— | Voltage RMS of U2 |
| ——— | Voltage RMS of U3 |

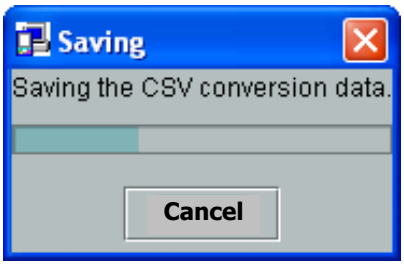
| | |
|---|---|
| Vertical axis zoom x25 | Change the vertical axis magnification rate by selecting from x1, x2, x5, x10, x25 or x50. |
| Interval 30 sec | Displays the recording interval setting on Model 3197. |
| Number of data 725 | Displays the number of data recorded by Model 3197. |

## 8. CSV Conversion Function

The measurement data of Model 3197 is recorded in the BINARY format.
"CSV (Comma Separated Value) Conversion Function" converts the measurement data recorded in Model 3197 to CSV text data for further processing on a Microsoft Excel spreadsheet or other similar application.

| | |
|---|---|
| CSV Save as CSV | Click this button to save the measurement data in CSV format. |




The CSV file will saved under a newly created folder named "CSV" in the same directory saved with the measurement data. The file name will be renamed with the extension "CSV". Settings and Event Monitor data cannot be converted to CSV.

| Measurement data (BINARY) | Content | CSV file |
|---|---|---|
| 97intvl.itv | TIMEPLOT Data (RMS, DIP/SWELL, Integrated Power) | 97intvl.csv |
| 97evt000.evt | Even waveforms data | 97evt000.csv |
| 97dv000.wdu | Event - Voltage Fluctuation Data | 97dv000.csv |
| 97inrush.inr | Event – Inrush Current Data | 97inrush.csv |
| 97demand.dem | Demand Graph Data | 97demand.csv |

The [Status] flag in the 97intvl.csv shows 32-bit width information.

| | | |
|---|---|---|
| 8bit | PLL | Unable to detect the zero-crossing. |
| 7bit | errUR | U1 is outside the set nominal voltage range. |
| 6bit | errf | Measurement value is outside the set frequency range. |
| 5bit | ovI3 | Over current (crest factor exceeded) I3 |
| 4bit | ovI2 | Over current (crest factor exceeded) I2 |
| 3bit | ovI1 | Over current (crest factor exceeded) I1 |
| 2bit | ovU3 | Over current (crest factor exceeded) U3 |
| 1bit | ovU2 | Over current (crest factor exceeded) U2 |
| 0bit | ovU1 | Over current (crest factor exceeded) U1 |

Channels not being used due to selected wiring will be displayed with "0" for data.

### 97intvl.csv Data-header Composition

| Classification | Header | Explanation |
|---|---|---|
| Date Time Status | Date | Date |
| | Time | Time |
| | Status | Status |
| TIME PLOT DIP/SWELL | Umax1, Umax2, Umax3 | Voltage(1/2) Maximum value |
| | Umin1, Umin2, Umin3 | Voltage(1/2) Minimum value |
| TIME PLOT RMS Maximum Values | MaxFreq | Frequency |
| | MaxUrms1, MaxUrms2, MaxUrms3 | Voltage |
| | MaxU+peak1, MaxU+peak2, MaxU+peak3 | Voltage Waveform Peak+ |
| | MaxU-peak1, MaxU-peak2, MaxU-peak3 | Voltage Waveform Peak- |
| | MaxIrms1, MaxIrms2, MaxIrms3, MaxIrms4 | Current |
| | MaxI+peak1, MaxI+peak2, MaxI+peak3, MaxI+peak4 | Current Waveform Peak+ |
| | MaxI-peak1, MaxI-peak2, MaxI-peak3, MaxI-peak4 | Current Waveform Peak- |
| | MaxUave | Voltage for 3phase |
| | MaxIave | Current for 3phase |
| | MaxPsum, MaxSsum, MaxQsum | Active Power, Apparent Power, Reactive Power |
| | MaxPFsum, MaxUunb | Power Factor |
| | MaxUthd1, MaxUthd2, MaxUthd3 | Total Harmonic Voltage Distortion Ratio |
| TIME PLOT RMS Minimum Values | MinFreq | Frequency |
| | MinUrms1, MinUrms2, MinUrms3 | Voltage |
| | MinU+peak1, MinU+peak2, MinU+peak3 | Voltage Waveform Peak+ |
| | MinU-peak1, MinU-peak2, MinU-peak3 | Voltage Waveform Peak- |
| | MinIrms1, MinIrms2, MinIrms3, MinIrms4 | Current |
| | MinI+peak1, MinI+peak2, MinI+peak3, MinI+peak4 | Current Waveform Peak+ |
| | MinI-peak1, MinI-peak2, MinI-peak3, MinI-peak4 | Current Waveform Peak- |
| | MinUave | Voltage for 3phase |
| | MinIave | Current for 3phase |
| | MinPsum, MinSsum, MinQsum | Active Power, Apparent Power, Reactive Power |
| | MinPFsum, MinUunb | Power Factor |
| | MinUthd1, MinUthd2, MinUthd3 | Total Harmonic Voltage Distortion Ratio |
| TIME PLOT RMS Average Values | AveFreq | Frequency |
| | AveUrms1, AveUrms2, AveUrms3 | Voltage |
| | AveU+peak1, AveU+peak2, AveU+peak3 | Voltage Waveform Peak+ |
| | AveU-peak1, AveU-peak2, AveU-peak3 | Voltage Waveform Peak- |
| | AveIrms1, AveIrms2, AveIrms3, AveIrms4 | Current |
| | AveI+peak1, AveI+peak2, AveI+peak3, AveI+peak4 | Current Waveform Peak+ |
| | AveI-peak1, AveI-peak2, AveI-peak3, AveI-peak4 | Current Waveform Peak- |
| | AveUave | Voltage for 3phase |
| | AveIave | Current for 3phase |
| | AvePsum, AveSsum, AveQsum | Active Power, Apparent Power, Reactive Power |
| | AvePFsum, AveUunb | Power Factor |
| | AveUthd1, AveUthd2, AveUthd3 | Total Harmonic Voltage Distortion Ratio |
| TIME PLOT Energy Quadergy | WP+, WP- | Energy (Delivered, Received) |
| | WQ_LAG, WQ_LEAD | Quadergy (Lagging, Leading) |

# Chapter 3: Software Specifications

## 1. General Specifications

| | |
|---|---|
| Computer: | PC AT compatible (CPU: 1GHz or faster) |
| Operating system: | Windows2000/XP (English or Japanese version) |
| Resolution: | 1024 x 768 dots, 16-bit color or higher |
| Memory: | 128MB or more (256MB or more recommended) |
| HDD: | Free disk space of 128MB or more |
| Connectable model: | 3197 POWER QUALITY ANALYZER |
| Connection method: | USB Ver.1.1 or 2.0 (full speed) |
| Identification: | Serial number (with name assignment function) |
| Product name: | 3197 Applications<br>(3197Communicator, 3197DataViewer) |

## 2. Function Specifications of 3197 Communicator

### -1. Data Transfer Function

| | | |
|---|---|---|
| Functions: | Download and delete | |
| Loadable data: | Measurement data (binary), Image data (BMP) | |
| Loading method: | Folder unit | |
| Measurement data: | SET file: | Setting data |
| | ITV file: | TIME PLOT data (RMS, DIP/SWELL, Integrated power) |
| | EVT file: | Event data (List, Waveform) |
| | WDU file: | Event voltage fluctuation data |
| | INR file: | Event inrush current data |
| | DEM file: | Demand graph data |
| | MON file: | Event monitor data |

### -2. Remote Operation Function

| | |
|---|---|
| Functions: | Display the same screen as the display on Model 3197 (with automatic refresh).<br>Operate Model 3197 using the virtual keys. |

## 3. Function Specifications of 3197 Data Viewer

### -3. Viewer Function

| | |
|---|---|
| Display screens: | Event list screen<br>Event details screen<br>Event waveform screen, Event voltage fluctuation graph screen, Event inrush current graph screen<br>TIME PLOT screen (RMS, DIP/SWELL) |

### -4. CSV Conversion Function

| | | |
|---|---|---|
| Functions: | Convert binary measurement data to CSV format. | |
| Conversion data: | ITV file: | TIME PLOT data (RMS, DIP/SWELL, Integrated power) |
| | EVT file: | Event data (List, Waveform) |
| | WDU file: | Event voltage fluctuation data |
| | INR file: | Event inrush current data |
| | DEM file: | Demand graph data |

HIOKI 3197Applications
Instruction Manual

Publication date: January 2007 Revised edition 1
Edited and published by HIOKI E.E. CORPORATION
Technical Support Section
All inquiries to International Sales and Marketing Department
81 Koizumi, Ueda, Nagano, 386-1192, Japan
TEL: +81-268-28-0562
FAX: +81-268-28-0568
E-mail: os-com@hioki.co.jp
URL http://www.hioki.co.jp/

Printed in Japan 3197A986-01

• All reasonable care has been taken in the production of this manual, but if you find any points which are unclear or in error, please contact your supplier or the International Sales and Marketing Department at HIOKI headquarters.
• In the interests of product development, the contents of this manual are subject to revision without prior notice.
• Unauthorized reproduction or copying of this manual is prohibited.

HIOKI

HIOKI E. E. CORPORATION

**HEAD OFFICE**
81 Koizumi, Ueda, Nagano 386-1192, Japan
TEL +81-268-28-0562 / FAX +81-268-28-0568
E-mail: os-com@hioki.co.jp / URL http://www.hioki.co.jp/

**HIOKI USA CORPORATION**
6 Corporate Drive, Cranbury, NJ 08512, USA
TEL +1-609-409-9109 / FAX +1-609-409-9108